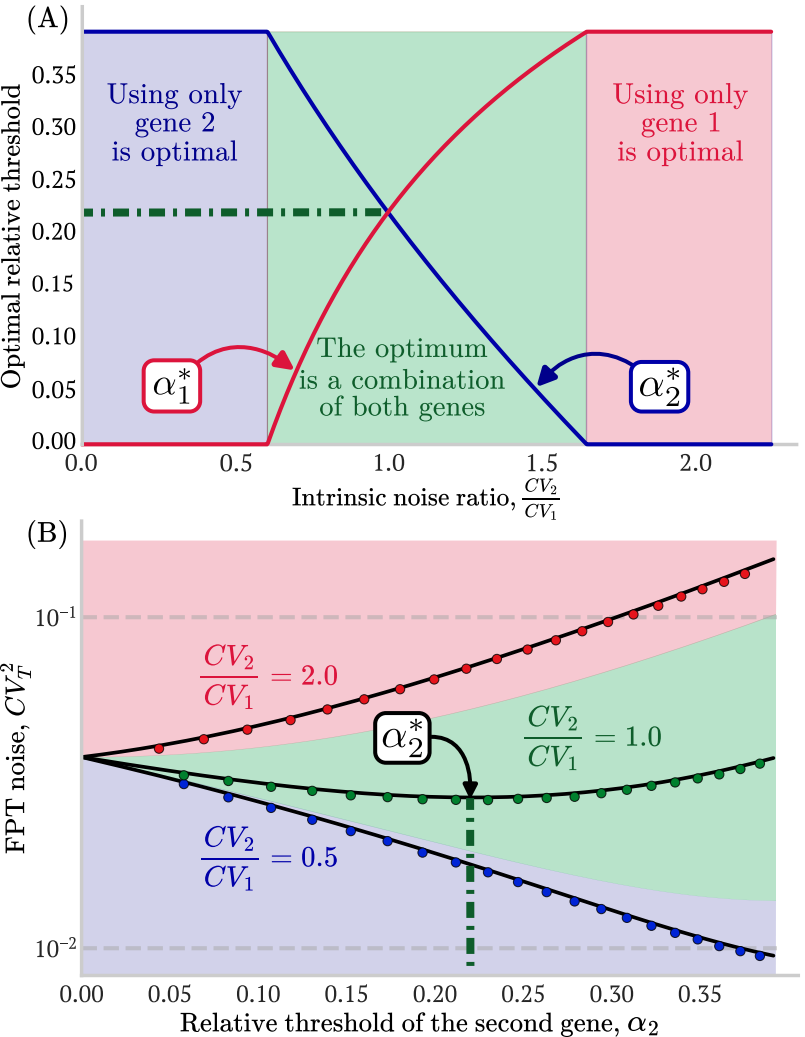
(A)
Optimal relative threshold
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Using only gene 2 is optimal
Using only gene 1 is optimal
The optimum is a combination of both genes
α_1^*
α_2^*
0.0
0.5
1.0
1.5
2.0
Intrinsic noise ratio, $\frac{CV_2}{CV_1}$
(B)
FPT noise, CV_T^2
10^{-1}
10^{-2}
$\frac{CV_2}{CV_1} = 2.0$
$\frac{CV_2}{CV_1} = 1.0$
$\frac{CV_2}{CV_1} = 0.5$
α_2^*
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Relative threshold of the second gene, α_2